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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: R26-DS-004

Project Title: An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka

	Identification and Analysis
Data Privacy Concerns	• The TAX Advisory system handles sensitive financial and personal information, including bank transaction records, income details, tax calculations, and individual taxpayer profiles. The Transaction Semantic Reasoning component processes bank statement data, while the Optimization and Explainable Engine and the Personalised Recommendation and Predictive Impact Modelling Engine use taxpayer information to support tax calculations and personalised recommendations. Therefore, protecting the confidentiality of this information is essential throughout the system. Any unauthorised access or disclosure could expose individuals' financial circumstances and affect their privacy. • To minimise privacy risks, anonymised or synthetic data should be used wherever possible during system development, testing, and evaluation. The individual component theses describe the use of synthetic taxpayer profiles and test cases to evaluate certain components. If real taxpayer information is required, it should be collected and processed only for clearly defined purposes, with informed consent obtained where necessary. Names, bank account numbers, and other identifying details should be removed from research datasets wherever possible. Data should only be retained for as long as necessary and securely deleted when it is no longer required. • Concrete privacy safeguards should be applied throughout the system. Access to taxpayer information should be restricted through role-based access control, ensuring that users can only view the information required for their responsibilities. Sensitive data should be protected using encryption during transmission and

	storage, while strong authentication should help prevent unauthorised access. Audit logs should record access to sensitive information and important data-related activities. In addition, personal financial information should not be unnecessarily included in application logs, error messages, or AI prompts sent to external services. Information shared between system components should be limited to the minimum required to perform each task. • Another important privacy concern is how taxpayer information is used by artificial intelligence models and external services. Financial information should not be used to train or improve AI models without an appropriate legal basis and proper authorisation. The system should avoid exposing personal details when generating tax explanations or personalised recommendations. Privacy risks should also be reviewed whenever new features, data sources, or integrations are introduced. Clear privacy information should be provided to users so they understand what information is collected, why it is needed, how it is processed, and how it is protected. These measures can help reduce the risk of data leaks, misuse, and unauthorised disclosure while maintaining taxpayer confidentiality and building trust in the TAX Advisory system.
Data Security	• The TAX Advisory system processes sensitive information such as bank transaction records, income details, taxpayer profiles, tax calculations, and recommendations. These data must be protected throughout the development and operation of the system to prevent unauthorised access, accidental disclosure, or modification. Access to financial records and system resources should be restricted to authorised users according to their roles and responsibilities. Development files, datasets, and model artefacts should be stored in secure, access-controlled locations rather than publicly accessible repositories.

	• Personal and financial information should be protected using encryption during storage and transmission wherever supported. Strong authentication and role-based access control should be used to prevent unauthorised users from accessing taxpayer information. Application logs and error messages should not expose sensitive details such as bank account numbers, personal identifiers, or financial records. API keys, database credentials, and other secrets should be stored securely in environment variables or an appropriate secrets-management solution rather than being hard-coded into the source code. • Since the system consists of several interconnected components, including Transaction Semantic Reasoning, the Optimization and Explainable Engine, the Intelligent Tax Advisory Language Model, and the Personalised Recommendation and Predictive Impact Modelling Engine, data exchanged between these components should be carefully controlled. Each component should receive only the information necessary to perform its assigned task. API endpoints should be protected through appropriate authentication and authorisation mechanisms, and input validation should be applied to reduce the risk of malicious requests or unauthorised data modification. Security checks should also be performed when integrating external AI services or other third-party tools. • Data integrity and availability should also be maintained throughout the research and development process. Source code, configuration files, datasets, and trained-model artefacts should be version-controlled and backed up in secure locations to prevent accidental loss or unauthorised changes. Access permissions, shared repository links, database connections, and backup locations should be reviewed regularly. Only the data and files necessary for the project should be retained, and obsolete copies should be securely deleted. Before

	deployment, the system should undergo security testing to identify weaknesses in authentication, access control, API protection, and data handling. Any security measures that have not been implemented or tested should be recognised as limitations of the current implementation rather than assumed to be effective.
Bias and Fairness	• The TAX Advisory system may be affected by biases in the financial data, tax records, and machine learning models used by its different components. For example, bank transaction descriptions may vary between banks, account types, and individuals, which could cause the Transaction Semantic Reasoning component to classify similar transactions differently. Similarly, the Personalised Recommendation and Predictive Impact Modelling Engine uses predictive models to rank tax-saving strategies and estimate their likelihood of adoption. If the training data does not adequately represent different taxpayer profiles and financial circumstances, the system may produce recommendations that are more suitable for some users than others. • To reduce these risks, the datasets used for training and evaluation should be reviewed for class imbalances, missing information, and insufficient representation of different financial situations. Synthetic taxpayer profiles can support testing, but they may not fully reflect the complexity of real-world taxpayer behaviour. Therefore, model performance should be evaluated across different income levels, employment types, transaction patterns, and taxpayer circumstances rather than relying only on overall accuracy. The system should also distinguish between identifying a financial transaction and determining its actual tax treatment, as an incorrect classification should not automatically lead to an unfair tax decision.

	• Fairness should also be considered when generating personalised recommendations and tax explanations. The system should provide recommendations based on relevant financial information and applicable tax rules rather than making assumptions about individuals based on unrelated personal characteristics. The Optimization and Explainable Engine supports transparency by linking tax calculations to relevant legal sources, while the advisory component is designed to provide evidence-based explanations. Uncertain classifications, unsupported recommendations, and potentially incorrect outputs should be flagged for further review where appropriate. Regular testing, human oversight, and updates to training data and tax rules can help identify potential biases, improve consistency, and ensure that the system provides fairer and more reliable support to different taxpayers. • Another important consideration is that biased or incomplete data can affect users' access to suitable tax-saving opportunities. For example, taxpayers with less common income sources or financial circumstances may receive less relevant recommendations if those situations are not adequately represented in the training data. Similarly, predictions about whether a person is likely to adopt a recommendation should not be treated as a measure of their eligibility or willingness to comply with tax regulations. The system should clearly communicate the limitations of its predictions, allow users or tax advisers to review questionable results, and avoid presenting automated recommendations as final decisions. These practices can help reduce unfair outcomes and ensure that the system supports taxpayers with different financial backgrounds more consistently.
Transparency and Accountability	• Transparency is important in the TAX Advisory system because its outputs may influence taxpayers' financial decisions and understanding of their tax obligations. The system should clearly explain how financial transactions are classified, how tax calculations are performed, and why particular tax-saving strategies are recommended. The Transaction Semantic Reasoning component

	identifies the likely meaning of bank transactions, while the Optimization and Explainable Engine provides tax calculations supported by relevant legal sources. The advisory component is designed to provide explanations based on retrieved legal evidence. Making this information available helps users understand how the system reaches its conclusions rather than relying entirely on automated results. • Clear documentation should be maintained throughout the development and evaluation of the system. This includes recording the datasets used, model configurations, tax rules, evaluation methods, and the assumptions made during implementation. The Optimization and Explainable Engine uses source verification and year-specific tax rules to improve traceability, while the personalised recommendation component uses ranking methods, predictive models, and explanations to support its recommendations. These methods should be documented so that the results can be reviewed, reproduced, and understood by other researchers. Any changes to tax rules, models, or system configurations should also be recorded to help identify the reasons for differences in results. • Accountability is equally important when the system produces incorrect classifications, outdated tax information, or unsuitable recommendations. Machine learning predictions should not be treated as guaranteed outcomes, and users should be informed that the system provides decision support rather than replacing qualified tax professionals. The system should identify uncertain results where possible and allow questionable outputs to be reviewed before users rely on them for important financial decisions. Researchers and developers are responsible for evaluating the system carefully, investigating errors, and reporting limitations honestly. In particular, the use of synthetic taxpayer data and limited evaluation datasets should be acknowledged when interpreting the research findings.

	• Furthermore, the ethical responsibilities of the research team include ensuring that the system's capabilities are not overstated and that its outputs remain traceable to the methods and evidence used. Evaluation results should be reported accurately, including unsuccessful tests, known weaknesses, and situations in which the system may provide unreliable information. Clear responsibilities should be established for reviewing tax-rule updates, maintaining the system, and addressing reported errors. Although explainability methods and documentation can improve transparency, they do not guarantee that every prediction or recommendation is correct. By maintaining proper records, communicating limitations, and ensuring appropriate human oversight, the research team can promote responsible use of the TAX Advisory system and maintain users' trust in its results.
Impact on Stakeholders	• The TAX Advisory system has the potential to benefit individual taxpayers, small business owners, tax advisers, and researchers by making tax-related information easier to understand and helping users explore their tax obligations. The Transaction Semantic Reasoning component supports the interpretation of bank transactions, while the Optimization and Explainable Engine helps users understand tax calculations and their supporting legal provisions. The personalised recommendation component identifies relevant tax-saving strategies and estimates their potential financial impact. These capabilities may reduce the effort required to organise financial information and help users make more informed decisions. However, the benefits depend on the accuracy of the information provided and the suitability of the recommendations for each taxpayer's circumstances. • The main stakeholders affected by the system include taxpayers, tax advisers, developers, researchers, and organisations responsible for managing tax-related information. Taxpayers may benefit from personalised guidance, but incorrect transaction classifications, outdated tax rules, or unsuitable recommendations could lead to misunderstandings and poor financial decisions.

	Tax advisers may find the system useful for reviewing financial information and explaining tax obligations, although they should still verify important outputs against applicable legislation. Developers and researchers are responsible for maintaining the system, protecting sensitive information, evaluating its performance, and addressing errors. The system should support users' decision-making rather than replace their judgement or remove their ability to seek professional advice. • The potential risks also differ depending on the stakeholder and the component involved. For example, an incorrect prediction from the Transaction Semantic Reasoning component could affect how a financial transaction is categorised, while an inaccurate legal explanation from the Intelligent Tax Advisory Language Model could cause a taxpayer to misunderstand a tax obligation. Similarly, recommendations from the Personalised Recommendation and Predictive Impact Modelling Engine may not reflect every individual's financial circumstances, particularly because some evaluations rely on synthetic data and proxy labels. The current research findings do not establish that the complete system is suitable for every real-world taxpayer situation. Therefore, users should be informed of these limitations, uncertain results should be reviewed where necessary, and tax calculations and legal explanations should be checked before important decisions are made. • In addition, the system must consider the privacy, fairness, and autonomy of the people whose financial information it processes. Taxpayers should understand how their information is used and should not be pressured to follow automated recommendations without considering their own circumstances. Unequal data representation or inaccurate predictions could also result in some users receiving less relevant guidance than others. To address these concerns, the research team should evaluate the system across different taxpayer profiles, maintain appropriate data protection measures,

	document known limitations, and provide clear explanations for important outputs. Before wider practical use, further validation with representative data, expert review of tax-related outputs, and security and fairness assessments would be necessary. These measures can help ensure that the system benefits its intended users while reducing potential harm to taxpayers and other stakeholders.